JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

CHAPTER 1: INTRODUCTION TO LINUX AND UBUNTU

COMMANDS, DEFINITIONS,AND OPTIONS,

1. whoami (Who Am I)

Definition:

The whoami command displays the username of the currently logged-in user.

Options:

No specific option is shown for this command in the chapter.



2. Working Directory pwd (Print)

Definition:

The pwd command displays the full path of the current working directory.

Options:

No specific option is shown for this command in the chapter.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

3. **uname (Unix Name)**

Definition:

The uname command displays information about the operating system and kernel..

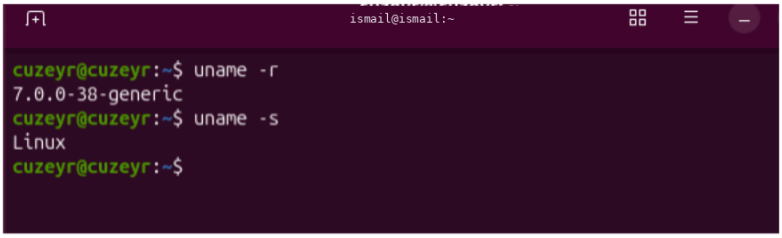
Options:

-r (Release):

Displays the current kernel release version.

-s (System Name):

Displays the operating system's kernel name.



4. **cat (Concatenate)**

Definition:

The cat command displays the contents of a file in the terminal.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

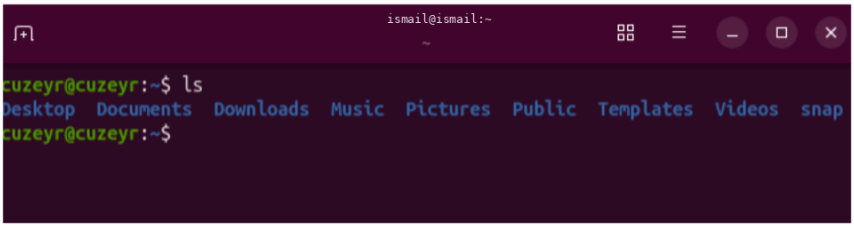
5. ls (List)

Definition:

The ls command displays the files and directories in the current directory.

Options:

No specific option is shown for this command in the chapter.



6. c (Copy)

Definition:

The cp command copies files or directories from one location to another.

Options:

No specific option is shown for this command in the chapter.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

7. **mv (Move)**

Definition:

The mv command moves files or directories from one location to another.

Options:

No specific option is shown for this command in the chapter.



8. **apt (Advanced Package Tool)**

Definition:

APT is a package management tool used in Debian-based Linux distributions, including Ubuntu.



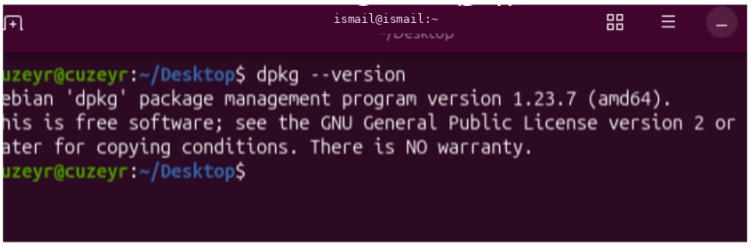
9. dpkg (Debian Package Manager)

Definition:

dpkg is a package management tool used to install, manage, and remove Debian software packages.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment



10. **dnf (Dandified YUM)**

Definition:

DNF is a package management tool used in Red Hat-based Linux distributions.

11. **zypper**

Definition:

Zypper is a command-line package management tool used in SUSE Linux distributions.

12**. yum (Yellowdog Updater, Modified)**

Definition:

YUM is a package management tool used in Red Hat-based Linux distributions.

Explanation:

It helps users install and manage software packages.

.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

CHAPTER 2 – Linux System Architecture, Command Line, and File Management

1. Commands, definitions, options, and examples

1. pwd (Print Working Directory)

Definition: Displays the full path of the current directory.

Options: No specific [illegible] apter.



2. cd (Change Directory)

Definition: Changes the current working directory.

Options:

.. = Move to the parent directory.

~ = Go to the home directory.

- = Return to the previous directory.

Example: cd /etc



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

3. **ls (List)**

Definition: Displays files and directories.

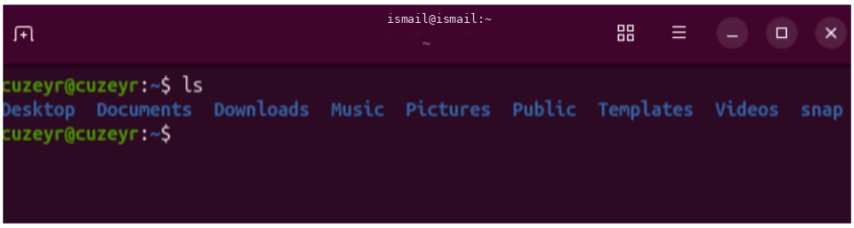
Options:

-a (All) = Shows hidden files.

-l (Long Listing) = Shows detailed information.

-h (Human-readable) = Displays readable file sizes.

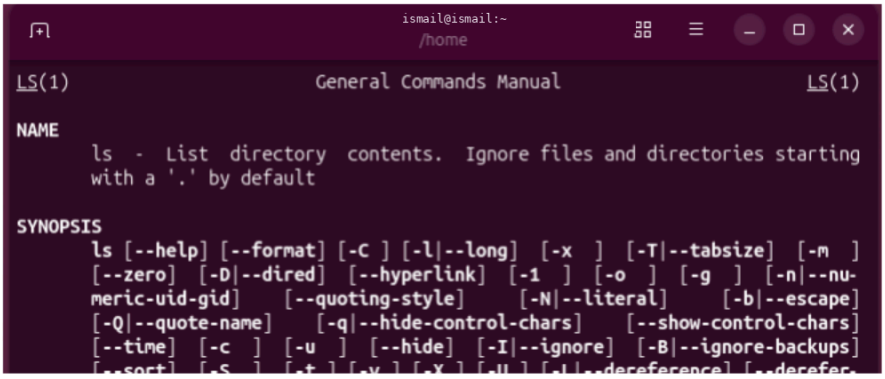
Example: ls –lah



4. **man (Manual)**

Definition: Displays the complete manual for a command.

Options: The command name is supplied as an argument.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

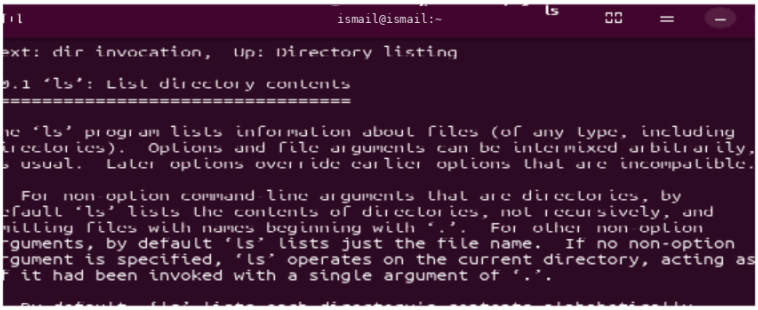
ID Number: C6240085

Class Name: CNS241

Assignment

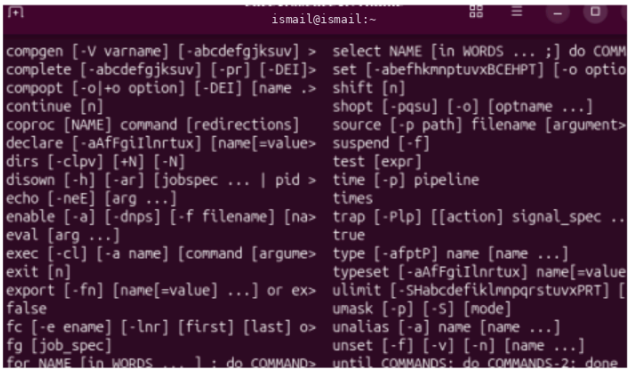
5. info (Information)

Definition: Displays detailed documentation for GNU commands.



6. help

Definition: Displays help for Bash built-in commands.



7. history

Definition: Displays previously executed commands.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

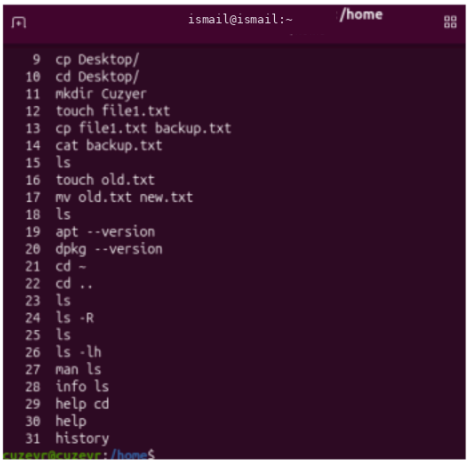
Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

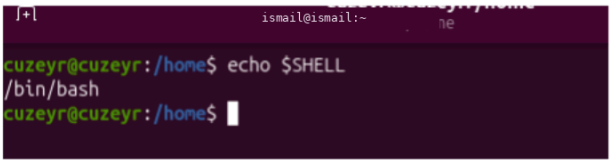
Class Name: CNS241

Assignment



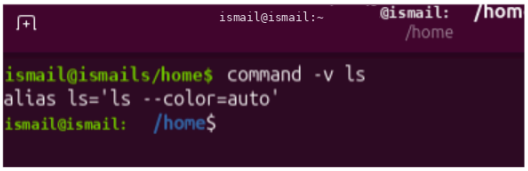
8. **echo**

Definition: Displays text or variable values.



9. **command -v**

Definition: Shows how the shell identifies or locates a command.

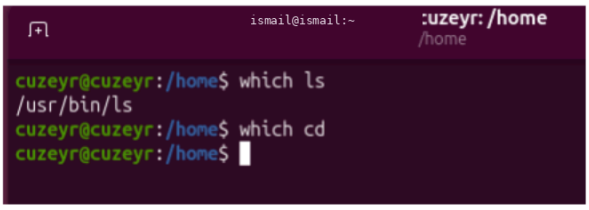


10. **which**

Definition: Searches the PATH directories for an executable command.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

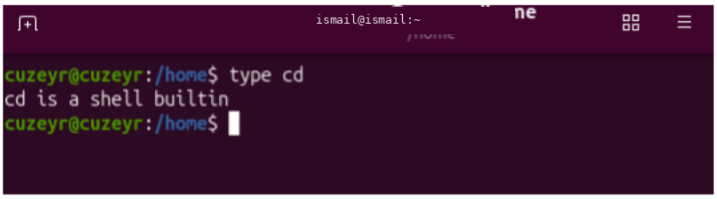
Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment



11. **type**

Definition: Identifies whether a command is a built-in, executable, alias, or function.

Example: type cd



12. **whereis**

Definition: Locates a command's executable, source, and manual files.

Example: whereis tar



13. **touch**

Definition: Creates an empty file or updates its timestamp.

Example: touch notes.txt

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

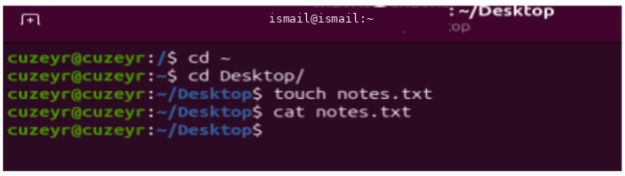
Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment



14. **mkdir (Make Directory)**

Definition: Creates a new directory.

Options:

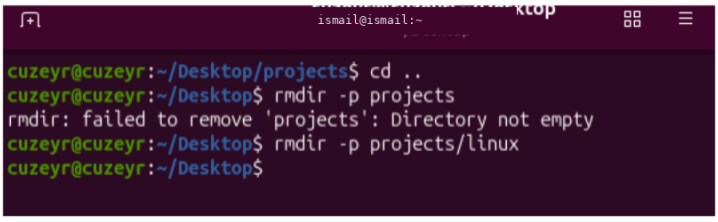
-p (Parents) = Creates missing parent directories.



15. **rmdir (Remove Directory)**

Definition: Removes an empty directory.

Example: rmdir week3



16. **cp (Copy)**

Definition: Copies files or directories.

Options:

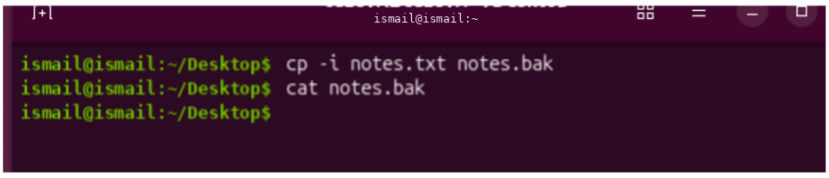
JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

-r (Recursive) = Copies directories and their contents.

-i (Interactive) = Asks before overwriting.

Example: cp -i notes.txt notes.bak

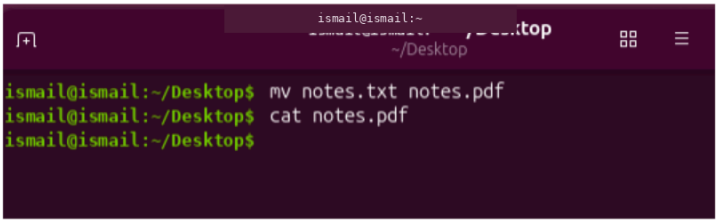


17. **mv (Move)**

Definition: Moves or renames files and directories.

Options:

-i (Interactive) = Asks before overwriting.



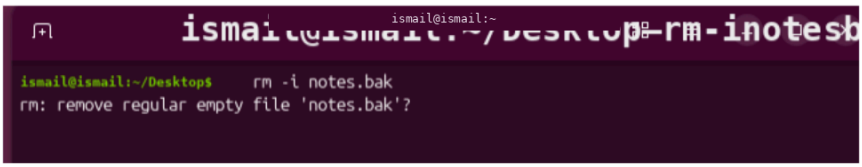
18. **rm (Remove)**

Definition: Removes files or directories.

Options:

-i (Interactive) = Asks before removing.

-r (Recursive) = Removes a directory and its contents.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

CHAPTER 3 – Text Processing, Searching, Archiving, and Editors

1. Commands, definitions, options, and examples

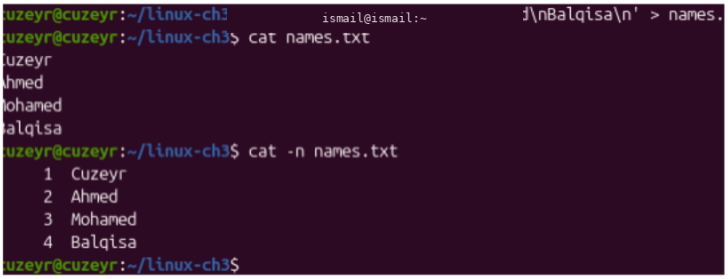
1. **cat (Concatenate)**

Definition: Displays file contents and combines files.

Options:

-n (Number) = Displays line numbers.

Example: cat -n names.txt



2. **less**

Definition: Displays long files one screen at a time.

Options: Use Space, b, /word, n, and q inside the viewer.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

3. **more**

Definition: Displays a file one screen at a time.



4. **head**

Definition: Displays the beginning of a file.

Options:

-n (Number) = Specifies the number of lines.



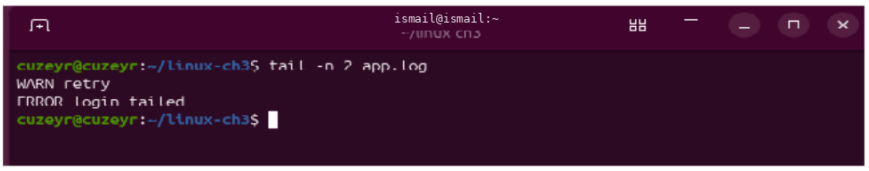
5. **tail**

Definition: Displays the end of a file.

Options:

-n (Number) = Specifies the number of lines.

-f (Follow) = Displays new lines as they are added.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

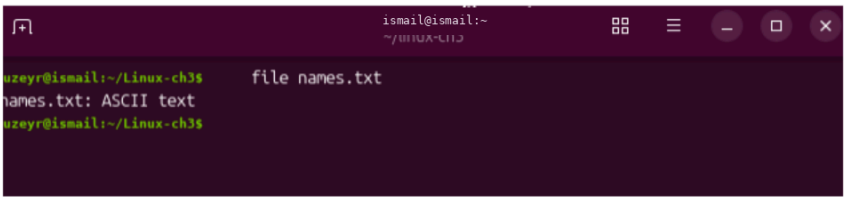
ID Number: C6240085

Class Name: CNS241

Assignment

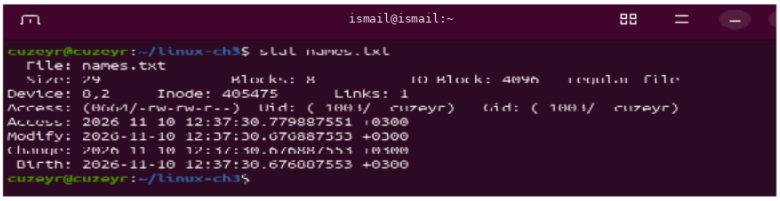
6. **file**

Definition: Identifies the actual type of a file.



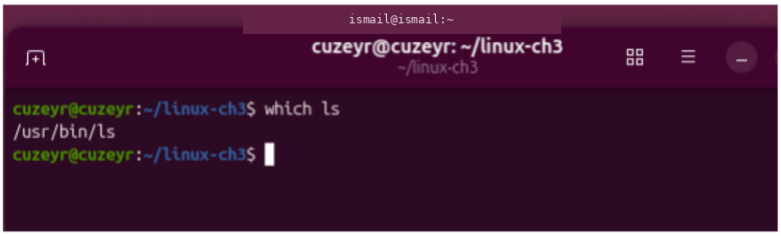
7. **stat (Status)**

Definition: Displays detailed file metadata.



8. **which**

Definition: Finds the location of an executable command.



9. **type**

Definition: Identifies a command's type.

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment



10. **command -v**

Definition: Shows how a command is resolved by the shell.



11. **whereis**

Definition: Locates executable files and manual pages.



12. **find**

Definition: Searches for files and directories.

Options:

-name = Searches by exact name or pattern.

-iname = Searches by name without case sensitivity.

-type f = Searches regular files.

-type d = Searches directories.

-size = Searches by file size.

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

-mmin = Searches by modification time in minutes.



13. **locate**

Definition: Finds files using a previously created filename database.



14. **updatedb (Update Database)**

Definition: Updates the database used by locate.

Note: May require administrator permission.

15. **grep (Global Regular Expression Print)**

Definition: Searches for matching text inside files.

Options:

-n (Line Number) = Shows matching line numbers.

-i (Ignore Case) = Ignores uppercase and lowercase differences.

-v (Invert Match) = Displays lines that do not match.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

-r (Recursive) = Searches inside subdirectories.

Example: grep -n 'ERROR' app.log



16. **sort**

Definition: Sorts lines of text.

Options:

-n (Numeric) = Sorts numbers numerically.

-r (Reverse) = Reverses the sorting order.



Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

17. **uniq (Unique)**

Definition: Removes adjacent duplicate lines.

Options:

-c (Count) = Counts repeated lines.

-d (Duplicates) = Displays duplicate lines.



18. **cut**

Definition: Extracts selected fields or characters.

Options:

-d (Delimiter) = Specifies the separator.

-f (Field) = Selects fields.

-c (Character) = Selects characters.

Example: cut -d: -f1 users.txt

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

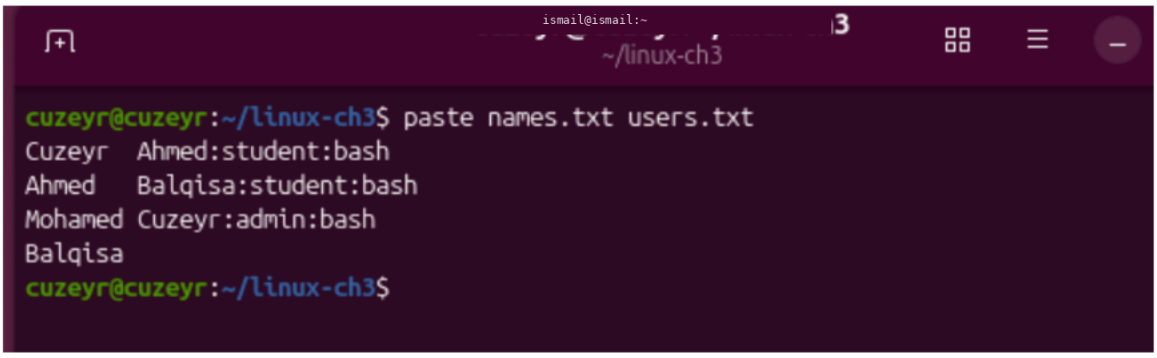
Class Name: CNS241

Assignment



19. **paste**

Definition: Combines lines from files side by side.



20. **wc (Word Count)**

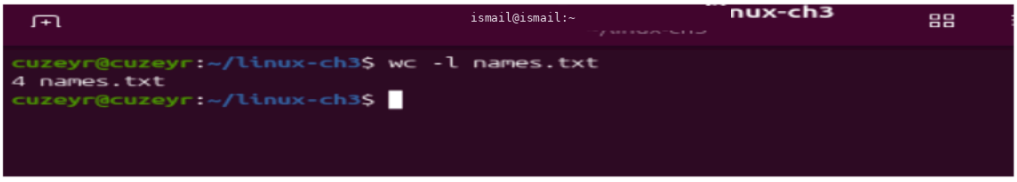
Definition: Counts lines, words, and characters or bytes.

Options:

-l (Lines) = Counts lines.

-w (Words) = Counts words.

-c (Bytes) = Counts bytes.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

21. **tr (Translate)**

Definition: Translates or deletes characters from standard input.

Options:

-c (Complement) = Selects characters not in the given set.

-s (Squeeze) = Replaces repeated characters with one.



22. **tar (Tape Archive)**

Definition: Creates, lists, and extracts archive files.

Options:

-c (Create) = Creates an archive.

-x (Extract) = Extracts an archive.

-v (Verbose) = Displays processed filenames.

-f (File) = Specifies the archive filename.

-t (List) = Lists archive contents.

-z (Gzip) = Uses gzip compression.

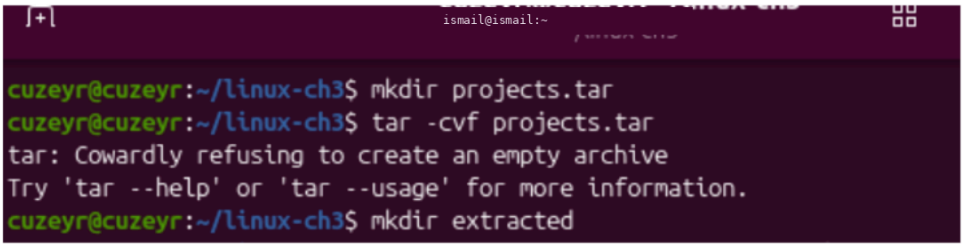
-j (Bzip2) = Uses bzip2 compression.

-J (XZ) = Uses xz compression.

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

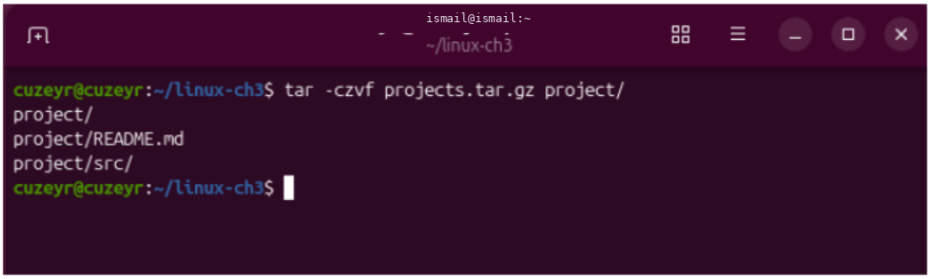
-C (Change Directory) = Specifies a directory.

Example: tar -cvf backup.tar project/



23. **gzip (GNU Zip)**

Definition: Compresses files.



24. **bzip2 (Block-Sorting File Compressor)**

Definition: Compresses files using bzip2 compression.

25. **xz (XZ Compression)**

Definition: Compresses files using XZ compression.

26. **zip**

Definition: Creates a ZIP archive.

27. **unzip**

Definition: Extracts ZIP archive contents.

Example: unzip backup.zip

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

28. **nano**

Definition: Opens a beginner-friendly text editor.

Shortcuts:

Ctrl+O = Save.

Ctrl+X = Exit.

Ctrl+W = Search.

Ctrl+K = Cut line.

Ctrl+U = Paste.



29. **vim (Vi Improved)**

Definition: Opens a text editor with different operating modes.

Example: vim notes.txt

30. **vi (Visual Editor)**

Definition: Opens the Vim editor on Ubuntu.

Example: vi notes.txt

31. **vimtut**

orDefinition: Opens a built-in Vim tutorial.

Example: vimtutor

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

32. **ln (Link)**

Definition: Creates a hard link by default.

33. **ln -s (Link Symbolic)**

Definition: Creates a symbolic link that stores a path.

Option:

-s (Symbolic) = Creates a symbolic link.

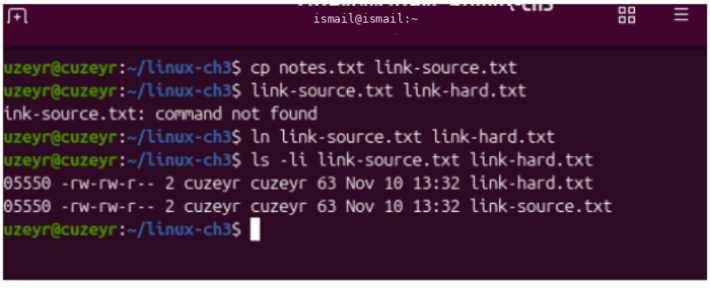
34. **ls -li**

Definition: Displays files with inode numbers.

Options:

-l (Long Listing) = Displays detailed information.

-i (Inode) = Displays the inode number.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

CHAPTER 4: PROCESS, SERVICE, AND SYSTEM MANAGEMENT

COMMANDS, DEFINITIONS, EXPLANATIONS, OPTIONS, AND ABBREVIATIONS

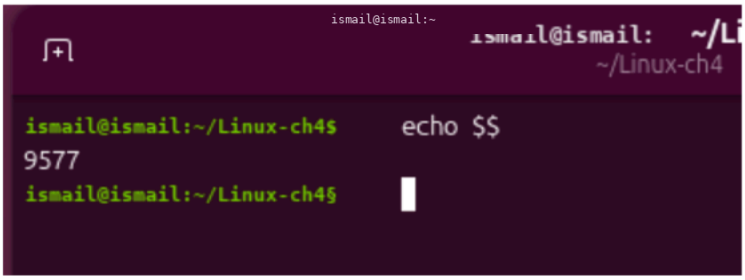
1. **Process**

Definition:

A process is a running program that is loaded into memory and assigned a unique identification number called PID.

$$:

A special shell variable that contains the PID of the current shell.



2**. ps (Process Status)**

Definition:

The ps command displays information about processes currently running in the system.

Options:

-f (Full Format):

Displays detailed information about processes, including the owner, PPID, and start time.

-p (Process ID):

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

Displays information about a specific process using its PID.

-o (Output Format):

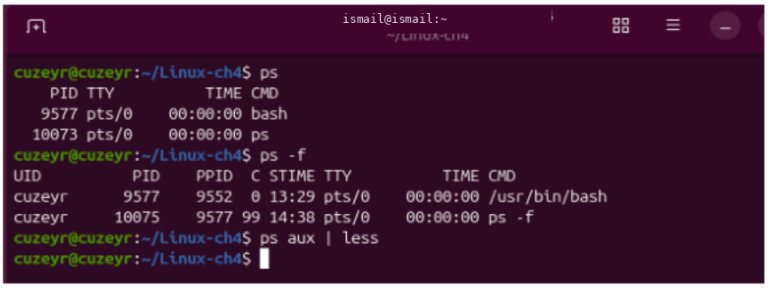
Allows users to select specific columns of process information.

-e (Every Process):

Displays every process in the system.

aux:

Displays processes belonging to all users, including CPU and memory usage.



3. **pstree (Process Tree)**

Definition:

The pstree command displays running processes in a tree structure.

Options:

-p (Process ID):

Displays process identification numbers in the process tree.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment



4**. top**

Definition:

The top command displays a live and continuously updated view of running processes and system resource usage.

Options and Keys:

-u (User):

Displays processes belonging to a specific user.

-p (Process ID):

Displays a specific process using its PID.

P (CPU Sorting):

Sorts processes according to CPU usage.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

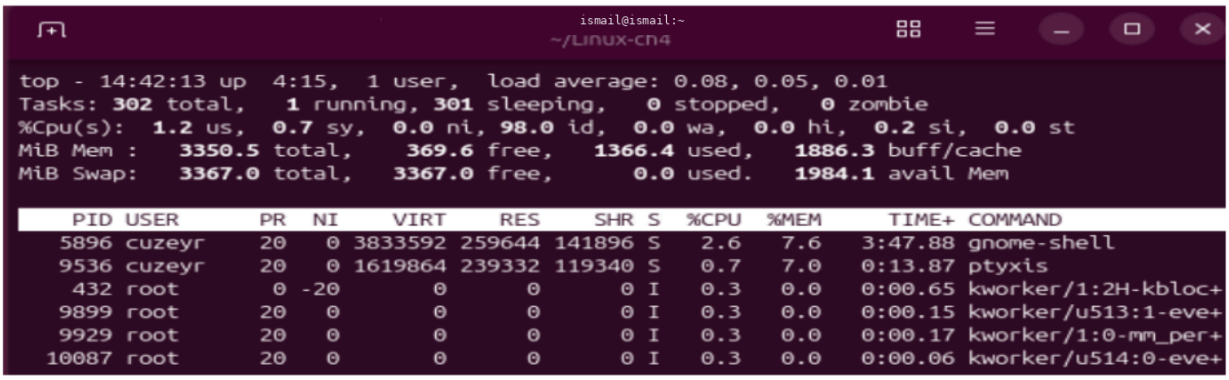
Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment



M (Memory Sorting):

Sorts processes according to memory usage.

k (Kill):

Allows the user to send a signal to a process.



r (Renice):

Allows the user to change the priority of a process.

q (Quit):

Exits the top command.

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

5. htop

Definition:

The htop command is an interactive process-monitoring tool that displays running processes with colours and additional controls.



command -v:

Checks whether a command is available and displays its location if installed.

6. **Foreground and Background Processes**

Definition:

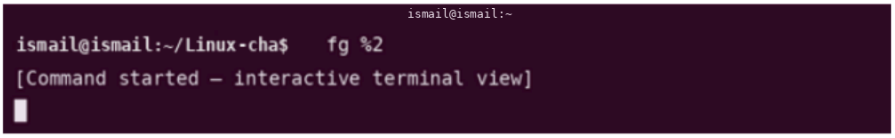
A foreground process is a process that runs directly in the terminal and holds the terminal until it finishes.



A background process is a process that runs without blocking the terminal.

Explanation:

Foreground processes prevent users from entering another command in the same terminal until the process finishes. Background processes allow users to continue working in the terminal while another process runs.



Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

Options and Symbols:

& (Ampersand):

Runs a command in the background.

Ctrl+C:

Sends the SIGINT signal to interrupt the foreground process.

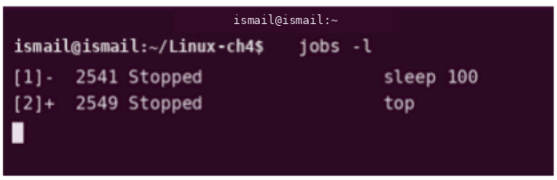
Ctrl+Z:

Sends the SIGTSTP signal to suspend the foreground process.

7. **jobs**

Definition:

The jobs command displays the jobs that are currently running or stopped in the current shell.



Options:

-l (Long Format):

Displays the process ID of each job together with its job number and status.

%N (Job Number):

Identifies a specific job using its job number.

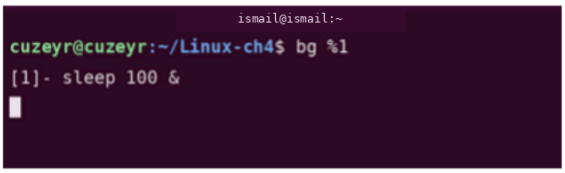
8**. bg (Background)**

Definition:

The bg command resumes a stopped job and allows it to continue running in the background.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment



Options:

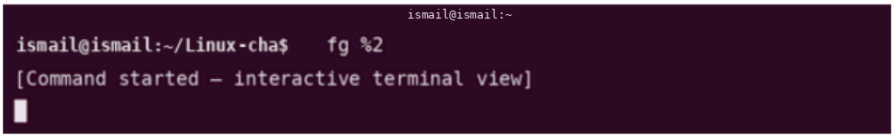
%N (Job Number):

Specifies the job number that should continue running in the background.

9**. fg (Foreground)**

Definition:

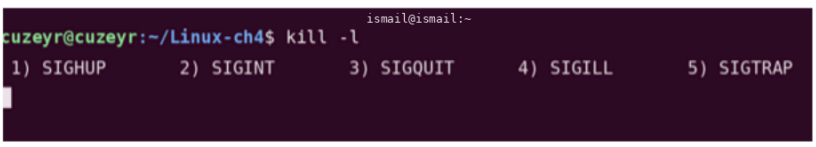
The fg command brings a background or stopped job to the foreground.



10**. Signals**

Definition:

A signal is a short message sent by the kernel to a process to request a specific action.



Important Signals:

SIGHUP (Signal Hang Up):

Signal number 1. It is commonly used by services to reload their configuration.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

SIGINT (Signal Interrupt):

Signal number 2. It interrupts a process, commonly through Ctrl+C.

SIGKILL (Signal Kill):

Signal number 9. It forcefully terminates a process without allowing normal cleanup.

SIGTERM (Signal Terminate):

Signal number 15. It politely requests a process to terminate normally.

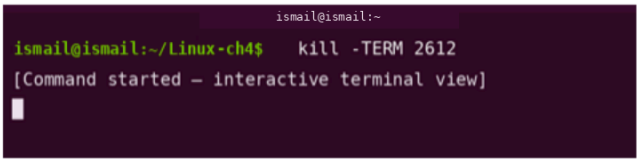
11. **kill**

Definition:

The kill command sends a signal to a process using its Process ID (PID).

Explanation:

It is used to request a process to terminate or perform another action depending on the signal sent. By default, it sends SIGTERM.



Options:

-TERM (Signal Terminate):

Sends the SIGTERM signal to request normal process termination.

-HUP (Signal Hang Up):

Sends the SIGHUP signal, commonly used to request a service to reload its configuration.

-9 (**SIGKILL):**

Forcefully terminates a process.

- **List Signals):**l (

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

Displays the available signal names.

%N (Job Number):

Identifies a job in the current shell.

12**. killall**

Definition:

The killall command sends a signal to processes that match a specified process name.

13**. pgrep (Process Grep)**

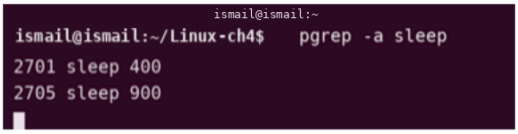
Definition:

The pgrep command searches for running processes by name or pattern and displays their process IDs.

Options:

-a (All Command Information):

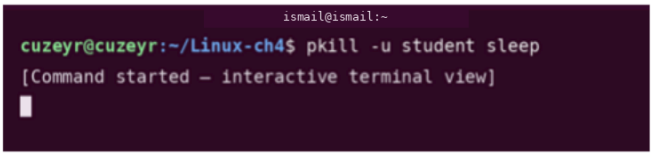
Displays the process ID and the full command line.



14. **pkill (Process Kill)**

Definition:

The pkill command sends a signal to processes by matching their names or patterns.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

15. **nice**

Definition:

The nice command starts a new process with a specified niceness value that affects its CPU scheduling priority.



Options:

-n (Niceness):

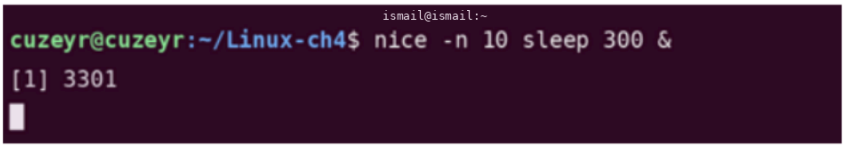
Specifies the niceness value for the new process.

Niceness Values:

* -20: Highest scheduling priority.

* 0: Default scheduling priority.

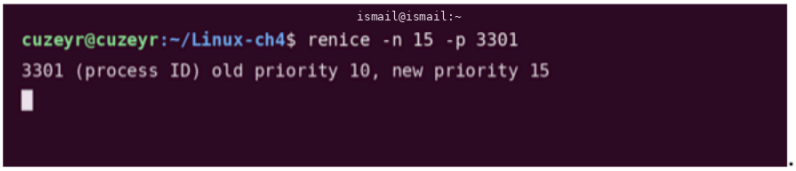
* 19: Lowest scheduling priority.



16. **renice**

Definition:

The renice command changes the niceness value of an already running process.



.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

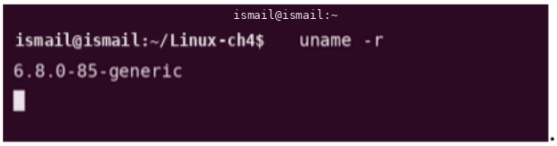
Class Name: CNS241

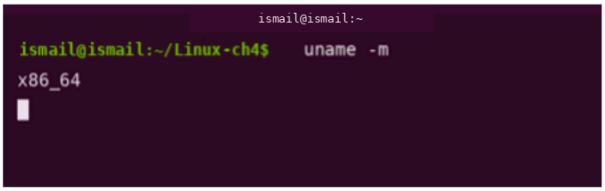
Assignment

17. **uname (Unix Name)**

Definition:

The uname command displays information about the running Linux kernel and machine architecture.

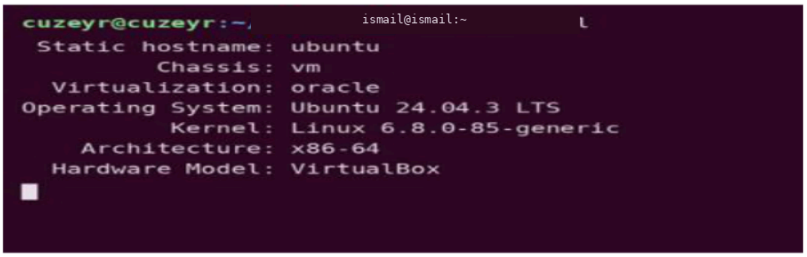




18. **hostnamectl**

Definition:

The hostnamectl command displays information about the system hostname, operating system, kernel, and hardware..



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

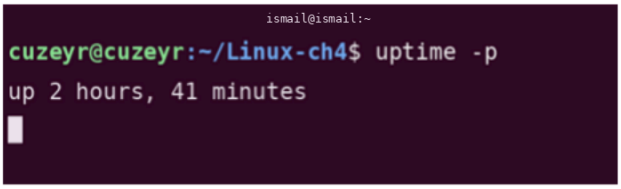
Class Name: CNS241

Assignment

19. **uptime**

Definition:

The uptime command displays how long the system has been running, the current time, the number of logged-in users, and the system load average.



20. **nproc (Number of Processors)**

Definition:

The nproc command displays the number of processing units available to the current process.



21. **free**

Definition:

The free command displays information about physical memory (RAM) and swap memory usage.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

22. **lscpu (List CPU)**

Definition:

The lscpu command displays detailed information about the CPU architecture and processor.

Options:

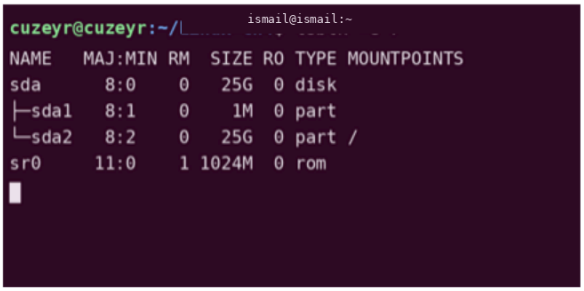
No specific option is shown for the main lscpu command in the chapter.



23. **lsblk (List Block Devices)**

Definition:

The lsblk command lists block devices available in the Linux system, including hard disks, partitions, and optical drives.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux

Prepared by: Ismail Adan Abdi

ID Number: C6240085

Class Name: CNS241

Assignment

24. /proc (Process Filesystem)

Definition:

The /proc directory is a virtual filesystem that provides information about running processes and the Linux kernel.

Important Files:

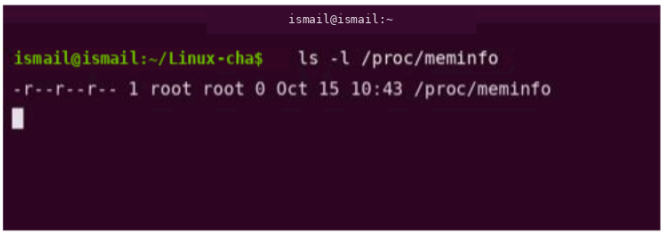
/proc/cpuinfo:

Contains information about the CPU.



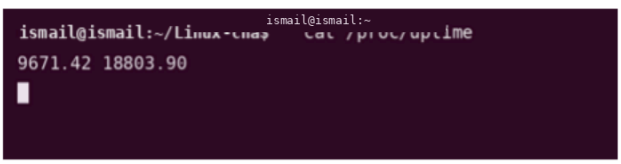
/proc/meminfo:

Contains information about memory usage.



/proc/uptime:

Contains information about how long the system has been running.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

/proc/loadavg:

Contains information about the system load average.



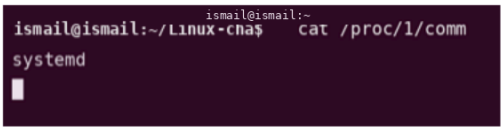
/proc/PID/status:

Contains information about a specific process, including its name, state, PID, and PPID.



/proc/PID/comm:

Contains the short command name of a process.



/proc/PID/cmdline:

Contains the command line used to start a process.

/proc/PID/exe:

A symbolic link to the executable file of a process.

25. **grep (Global Regular Expression Print)**

Definition:

The grep command searches for specific text or patterns inside files or command output.



Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

Options:

-m 1 (Maximum Matches):

Stops searching after finding one matching line.

-c (Count):

Counts the number of lines that match the specified pattern.

^ (Caret):

Represents the beginning of a line when used in a search pattern.

26. **/sys (System Filesystem)**

Definition:

The /sys directory is a virtual filesystem that provides information about hardware devices, device drivers, and kernel components.

Important Directories:

/sys/block:

Contains information about block devices.

/sys/class:

Contains devices grouped according to their type.

/sys/class/net:

Contains information about network interfaces.

27. **systemd**

Definition:

systemd is the system and service manager used by Ubuntu Linux.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

Important Terms:

PID 1:

The process identification number assigned to systemd.

.service:

The file extension commonly used for systemd service unit files.

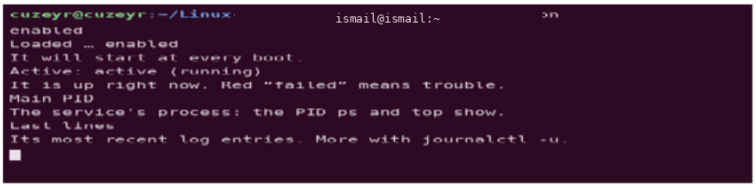
28. **systemctl (System Control)**

Definition:

The systemctl command is used to control and manage systemd services and units.

Explanation:

It allows administrators to check service status, start and stop services, restart services, and control whether services start automatically during boot.



Options and Actions:

status:

Displays the current status of a service, including whether it is running or enabled.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY



Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

is-active:

Checks whether a service is currently active or running.

is-enabled:

Checks whether a service is configured to start automatically during system boot.

start:

Starts a stopped service immediately.

stop:

Stops a currently running service immediately.

restart:

Stops and starts a service again.

reload:

Requests a running service to reload its configuration without stopping it. It is supported only by some services.

reload-or-restart:

Reloads a service if it supports reloading. Otherwise, it restarts the service.

enable:

Configures a service to start automatically during system boot.

disable:

Prevents a service from starting automatically during the next system boot.

–now:

Performs the service action immediately while also changing its automatic startup setting.

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

list-units:

Displays units currently loaded by systemd.

–type=service:

Displays service units only.

–state=running:

Displays services that are currently running.

–state=failed:

Displays services that have failed.

cat:

Displays the contents of a systemd unit file.

show:

Displays detailed properties of a systemd unit.

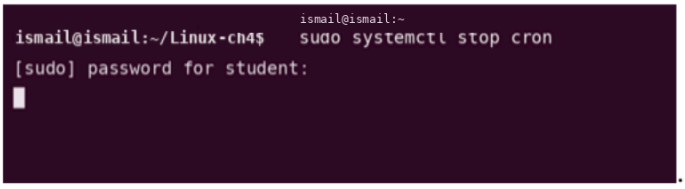
-p (Property):

Selects a specific property to display.

29. **sudo (Superuser Do)**

Definition:

The sudo command allows an authorized user to execute a command with administrator privileges.



Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

30. **journalctl (Journal Control)**

Definition:

The journalctl command displays system logs collected by the systemd journal.

Options:

-n (Number):

Displays the most recent specified number of log entries.

-b (Boot):

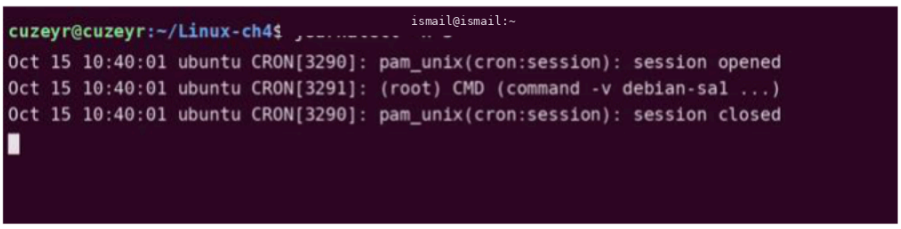
Displays log messages from the current system boot.

-p err (Priority Error):

Displays error-priority messages and more severe messages.

-u (Unit):

Displays log messages belonging to a specific service or systemd unit.



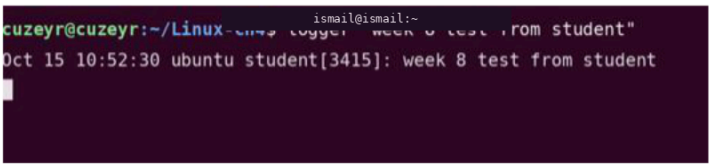
31. **logger**

Definition:

The logger command writes a message to the system log..

JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

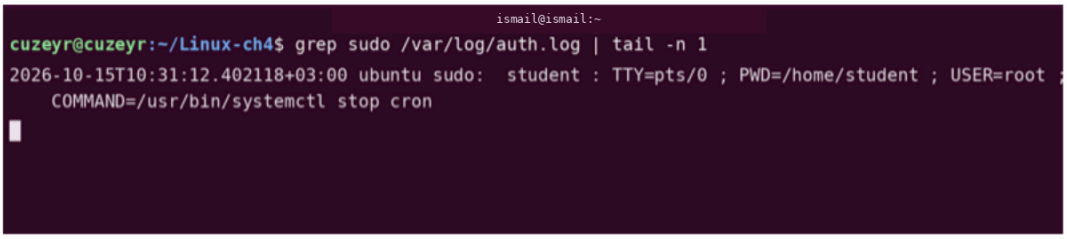
Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment



32. **/var/log Directory**

Definition:

The /var/log directory stores traditional text log files that record system activities and events.



Important Files:

/var/log/syslog:

Contains general system messages.

/var/log/auth.log:

Contains authentication-related information, including login, sudo, and SSH activities.

/var/log/kern.log:

Contains kernel-related messages.

/var/log/journal/:

Contains systemd journal data stored in binary format.



JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

33**. less**

Definition:

The less command displays the contents of a file one page at a time.

34. **tail**

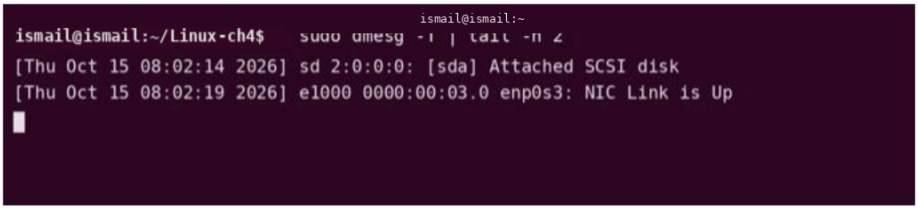
Definition:

The tail command displays the last lines of a file.

Options:

-n (Number):

Specifies the number of lines to display from the end of a file.



35. **last**

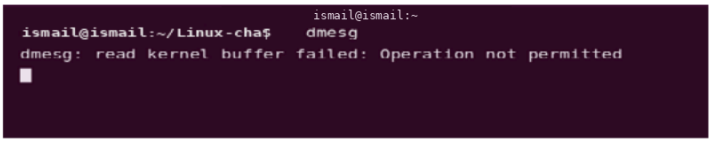
Definition:

The last command displays information about previous user logins and system reboots.

36. **dmesg (Display Message)**

Definition:

The dmesg command displays messages from the Linux kernel ring buffer.



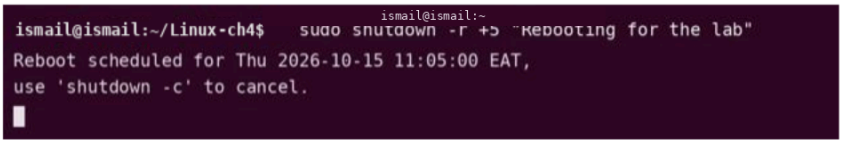
JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

37. **shutdown**

Definition:

The shutdown command safely shuts down or restarts the Linux system.



Options and Arguments:

now:

Schedules the shutdown or restart immediately.

+N:

Schedules the shutdown or restart after a specified number of minutes.

-r (Reboot):

Restarts the system instead of powering it off.

-c (Cancel):

Cancels a previously scheduled shutdown.

38. reboot

Definition:

The reboot command restarts the Linux system.



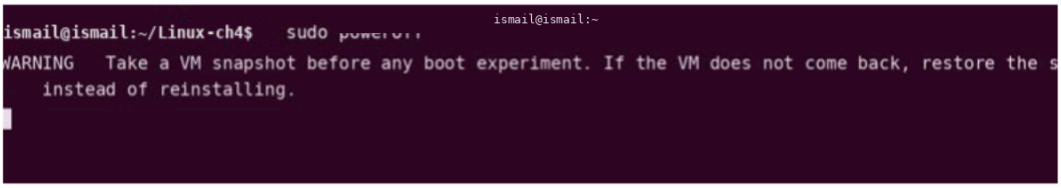
JAMHURIYA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Subject: Linux
Prepared by: Ismail Adan Abdi
ID Number: C6240085
Class Name: CNS241
Assignment

39. **poweroff**

Definition:

The poweroff command shuts down the Linux system and switches off the machine.



40**. sshd -t**

Definition:

The sshd -t command checks the SSH server configuration for syntax errors.

